



Gowin AXI-Stream FIFO IP

User Guide

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Revision History

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1 About This Guide

1.1 Purpose

The purpose of Gowin AXI-Stream FIFO IP User Guide is to help you to quickly understand the features and usage of Gowin AXI-Stream FIFO IP by providing the descriptions of the functions, signal definition, parameters, and reference design. The software screenshots and the supported products listed in this manual are based on Gowin Software 1.9.9 Beta-3. As the software is subject to change without notice, some information may not remain relevant and may need to be adjusted according to the software that is in use.

1.2 Related Documents

The latest user guides are available on the GOWINSEMI website. You can find the related documents at www.gowinsemi.com:

- [SUG100, Gowin Software User Guide](#)
- [DS102, GW2A series of FPGA Products Data Sheet](#)
- [DS226, GW2AR series of FPGA Products Data Sheet](#)
- [DS100, GW1N series of FPGA Products Data Sheet](#)
- [DS117, GW1NR series of FPGA Products Data Sheet](#)
- [DS821, GW1NS series of FPGA Products Data Sheet](#)
- [DS861, GW1NSR series of FPGA Products Data Sheet](#)
- [DS841, GW1NZ series of FPGA Products Data Sheet](#)
- [DS1228, Arora V FPGA Products Overview](#)
- [DS981, Arora V 138K & 75K FPGA Products Data Sheet](#)

1.3 Terminology and Abbreviations

The terminology and abbreviations used in this manual are as shown in Table 1-1.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Meaning	
ALU	Arithmetic Logic Unit	
AXI4	Advanced eXtensible Interface 4	
BSRAM	Block Static Random Access Memory	
FIFO	First In First Out	
I/O	Input/Output	
IP	Intellectual Property	
LUT	Look-up Table	
REG	Register	
M1	AXI4-Lite Master	Collectively referred to as M
M2	AXI4 Master	
A-S	AXI4-Stream Master/Slave	
TX-FIFO	FIFO for storing transactions from M→ IP	
RX-FIFO	FIFO for storing transactions from A-S→IP	
WE-ADDR	Write Valid Address for M→IP	
WE-DATA	Write Valid Transaction for M→IP	
WE-RESP	Write Valid Response for IP→M	
OE-ADDR	Read Valid Address for M→IP	
OE-DATA	Read Valid Transaction for M→IP	
TX-Sel	Transaction Selector for M→IP	
RX-Sel	Transaction Selector for IP→M	

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly using the information provided below.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Overview

2.1 Introduction

Gowin AXI-Stream FIFO IP enables the conversion between AXI4/AXI4-Lite transactions and AXI4-Stream transactions, allowing users to perform memory-mapped access on the AXI4-Stream interface using this IP.

Table 2-1 Gowin AXI-Stream FIFO IP Overview

Gowin AXI-Stream FIFO IP	
Logic Resource	See Table 2-2.
Delivered Doc.	
Design Files	Verilog (encryption)
Reference Design	Verilog
TestBench	Verilog
Test and Design Flow	
Synthesis Software	GowinSynthesis
Application Software	Gowin Software (V1.9.9Beta-3 and above)

Note!

For the devices supported, you can click [here](#) to get the information.

2.2 Features

- 32-bit AXI4-Lite monitoring interface
- Configurable transaction interface (AXI4 or AXI4-Lite). For AXI4-Lite, the transaction width is 32 bits. For AXI4, the transaction width is 32/64 bits.
- Independently configurable internal TX-FIFO and RX-FIFO
- Full duplex operation
- Provides interruption and monitoring
- Independently configurable transaction transmission mode (Store-and-Forward Mode/Cut-Through Mode)

2.3 Operating Frequency

The operating frequency of Gowin AXI-Stream FIFO IP is constrained by the clock provided by M1 and the maximum clock frequency supported by the selected FPGA. For example, when using the GW2A-55 series of devices, the default configuration allows a maximum operating frequency of 150 MHz.

2.4 Resource Utilization

Gowin AXI-Stream FIFO IP is implemented with Verilog HDL. Its performance and resource utilization may vary when the design is employed in different devices, or at different densities, speeds, or grades. Taking Gowin GW2A-55 device as an instance, the resource utilization is as shown in Table 2-2. For the resource utilization of other devices, please refer to later release information.

Table 2-2 Resource Utilization

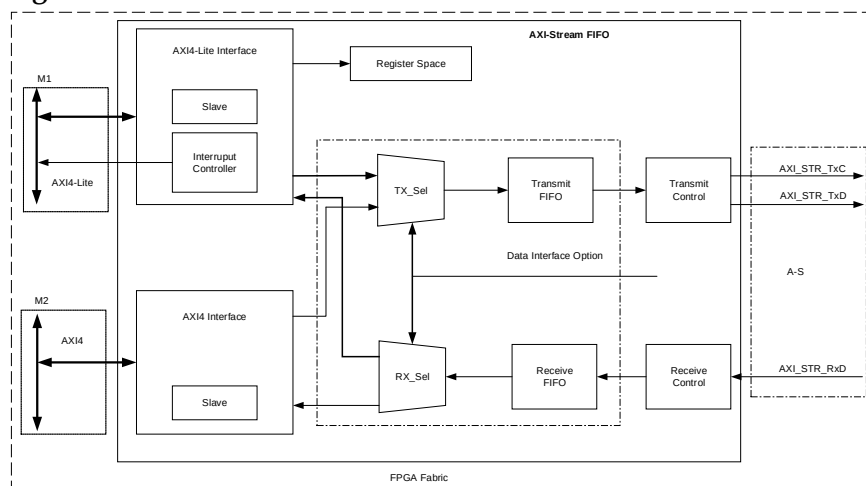
Device	Programming Language	LUTs	ALUs	REGs	BSRAM
GW2A-55	Verilog HDL	1863	28	845	2

3 Functional Description

3.1 System Block Diagram

Gowin AXI-Stream FIFO IP is composed of the AXI4-Lite Interface, AXI4 Interface, Data Interface, Transmit Control, and Receive Control, as shown in Figure 3-1. The IP externally connects to the Master port and AXI4-Stream port, supporting the AXI4 standard protocol.

Figure 3-1 Gowin AXI-Stream FIFO IP Structure



3.2 Modules

- **AXI4-Lite Interface:** Connects the IP to Master 1, enabling read and write of memory-mapped transactions (Slave) and monitoring transactions (Interrupt Controller, Register Space).
- **AXI4 Interface:** Connects the IP to Master 2, enabling read and write of memory-mapped transactions (Slave).
- **Data Interface:** TX FIFO stores and forwards transaction "M → IP → A-S", while the RX FIFO performs the reverse process.
- **Transmit Control/Receive Control:** Connects the IP and A-S to realize transmitting and receiving of stream transactions.
- **Register Space:** Monitors the IP, as detailed in section 3.3 Register Description.

3.3 Register Description

All registers in the Register Space are used for monitoring the IP, and each is 4 bytes in size. Specific names, addresses, and descriptions can be found in the Table 3-1.

Table 3-1 Description of Registers

REG	Address	Type	Description
ISR	0x00	Read/Write	Interrupt Status Register
IER	0x04	Read/Write	Interrupt Enable Register
TDFR	0x08	Write-only	Transmit Data FIFO Reset Register
TDFV	0x0C	Read-only	Transmit Data FIFO Vacancy Register
TLR	0x14	Write-only	Transmit Length Register
RDFR	0x18	Write-only	Receive Data FIFO Reset Register
RDFO	0x1C	Read-only	Receive Data FIFO Occupancy Register
RLR	0x24	Read-only	Receive Length Register
SRR	0x28	Read-only	AXI4-Stream Reset Register
TDR	0x2C	Write-only	Transmit Destination Register
RDR	0x30	Read-only	Receive Destination Register

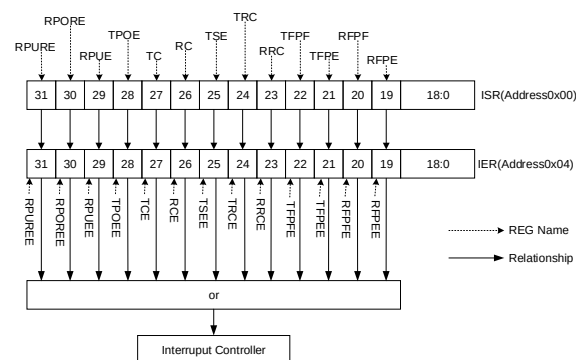
Note!

- M1 writes the WE-DATA with address 0x10 into the TX-FIFO.
- M1 reads the OE-DATA with address 0x20 from RX-FIFO.
- M2 writes the WE-DATA with address 0x00 into the TX-FIFO.
- M2 reads the OE-DATA with address 0x00 from RX-FIFO.

3.3.1 ISR and IER

Interrupts are managed by both the ISR and IER. When both ISR[n] and IER[n] are 1, the output interrupt signal is valid (Interrupt=1). The interrupt structure is as shown in Figure 3-2.

Figure 3-2 Interrupt Structure Diagram



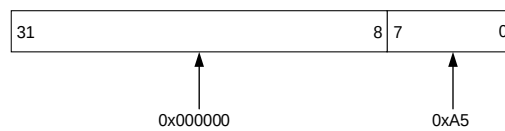
In the Figure 3-2, each bit in the ISR[31:19] represents an interrupt status, and IER[31:19] is used to enable/disable the corresponding ISR[31:19] bits. When ISR[n] is 1, it indicates that an interrupt has occurred or is ongoing. If IER[n] is also 1 at this time, the Interrupt=1 is output. The interrupt definition is as shown in Table 3-2.

Table 3-2 ISR Interrupt Definition

Bit	Name	Definition
18:0	Reserved	All defaults to 0, and there is no effect when write/read as 1.
19	RFPE	Receive FIFO Programmable Empty: Generated when the difference between the read and write pointers of the RX-FIFO reaches the programmable empty threshold.
20	RFPF	Receive FIFO Programmable Full: Generated when the difference between the read and write pointers of the RX-FIFO reaches the programmable full threshold.
21	TFPE	Transmit FIFO Programmable Empty: Generated when the difference between the read and write pointers of the TX-FIFO reaches the programmable empty threshold.
22	TFPF	Transmit FIFO Programmable Full: Generated when the difference between the read and write pointers of the TX-FIFO reaches the programmable full threshold.
23	RRC	Receive Reset Complete: Generated when the receive logic reset is complete.
24	TRC	Transmit Reset Complete: Generated when the transmit logic reset is complete.
25	TSE	Transmit Size Error: Generated when the number of bytes written to the TX-FIFO does not match the value written to the TLR.
26	RC	Receive Complete: Generated when at least one packet is successfully received and the received transaction and the read RLR value are available.
27	TC	Transmit Complete: Generated when at least one transmission is successful.
28	TPOE	Transmit Packet Overrun Error: Generated when the TX-FIFO is full and another transaction needs to be stored, requiring the transmit logic to be reset to recover.
29	RPUE	Receive Packet Underrun Error: Generated when the RX-FIFO is empty and another transaction needs to be read, requiring the receive logic to be reset to recover.
30	RPORE	Receive Packet Overrun Read Error: Generated when the RX-FIFO reads more words than those currently being processed within a packet, requiring the receive logic to be reset to recover.
31	RPURE	Receive Packet Underrun Read Error: Generated when attempting to read an empty RLR, requiring the receive logic to be reset to recover when transaction read is invalid.

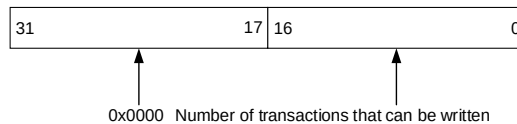
3.3.2 TDFR

When the value of TDFR is 0x000000A5, as shown in Figure 3-3, it generates a TX-FIFO reset signal. Writing other values has no effect, and 0x000000A5 is only written when TX-FIFO to A-S (AXI4-Stream) transactions are idle.

Figure 3-3 TDFR

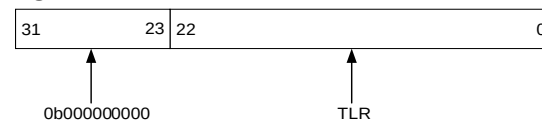
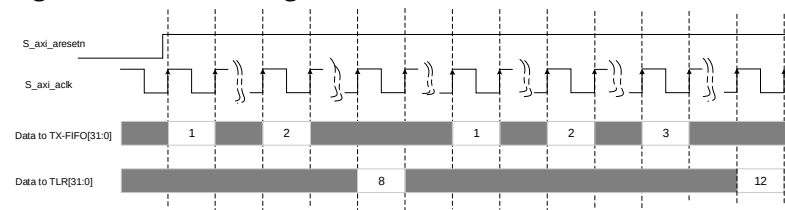
3.3.3 TDFV

TDFV is used to record the available space in the TX-FIFO, as shown in Figure 3-4.

Figure 3-4 TDFV

3.3.4 TLR

TLR is used to store the packet length value (in bytes) for each transaction, as shown in Figure 3-5. The IP first needs to completely write each transaction from the M into the TX-FIFO, and then write the corresponding transaction packet length value into the TLR, as shown in Figure 3-6.

Figure 3-5 TLR**Figure 3-6 TLR Timing**

- **Store-and-Forward Mode**

In this mode, each transaction from the M must be completely stored in the TX-FIFO. Only after a valid value is written to the TLR can the transaction be taken from the TX-FIFO and transmitted to the A-S.

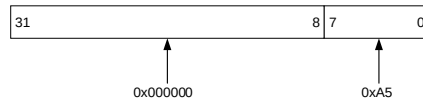
- **Cut-Through Mode**

In this mode, part of the transaction can be forwarded to the A-S without waiting for the integrity of the transaction packet (at least one cycle of the transaction is reserved in the TX-FIFO to wait for a valid value written to the TLR), and then the remaining at least one cycle of the transaction can be forwarded to the A-S after a valid value is written to the TLR.

3.3.5 RDFR

When the value of RDFR is 0x000000A5, as shown in Figure 3-7, it generates an RX-FIFO reset signal. Writing other values has no effect, and 0x000000A5 is only written when RX-FIFO to M transactions are idle.

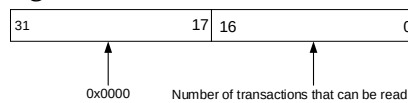
Figure 3-7 RDFR



3.3.6 RDFO

RDFO is used to record the utilized space in the RX-FIFO, as shown in Figure 3-8.

Figure 3-8 RDFO



3.3.7 RLR

RLR is used to store the transaction length value (in bytes) that the M currently needs to read.

- Store-and-Forward Mode

In this mode, each transaction from A-S must be completely stored in the RX-FIFO. The IP calculates the RLR value and waits for M1 to read it. Afterward, the M reads the transaction based on the RLR value, as shown in Figure 3-9 and Figure 3-10.

Figure 3-9 RLR (Store-and-Forward Mode)

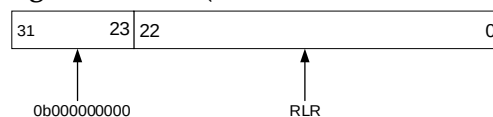
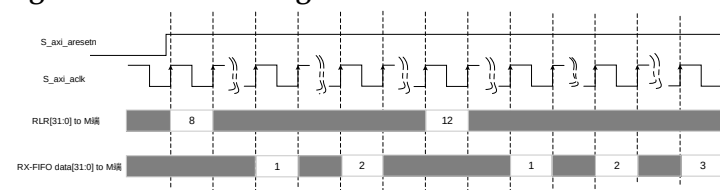


Figure 3-10 RLR Timing



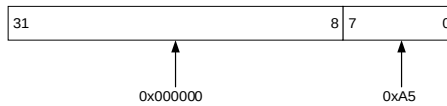
- Cut-Through Mode

In this mode, when the RX-FIFO is non-empty, the IP can set the highest bit of the RLR to 1. At this point, the M can begin reading transactions from the RX-FIFO without waiting for the entire transaction packet to be complete. Once the entire transaction packet has been read, the IP will set the highest bit of the RLR to 0, indicating that the transaction packet has been completely read, as shown in Figure 3-11.

Figure 3-11 RLR (Cut-Through Mode)

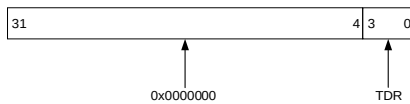
3.3.8 SRR

When the value of SRR is 0x000000A5, as shown in Figure 3-12, it generates an A-S reset signal. Writing other values has no effect.

Figure 3-12 SRR

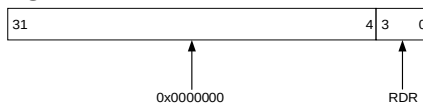
3.3.9 TDR

TDR is used to store the destination address for the transaction to be sent, as shown in Figure 3-13. It is not used in the current version.

Figure 3-13 TDR

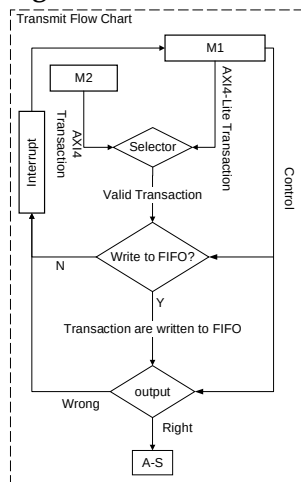
3.3.10 RDR

RDR is used to store the destination address for the transaction to be received, as shown in Figure 3-14. It is not used in the current version.

Figure 3-14 RDR

3.4 Operating Principles

3.4.1 Operating Principle for Transmitting

Figure 3-15 Transmit Flow Diagram

The Transmit flow diagram is as shown in Figure 3-15, and it works as follows:

When the interface is set to M1, the IP stores and forwards transactions from M1 to A-S.

1. M1→IP

The IP writes the WE-DATA with the address 0x10 from M1 into the TX-FIFO in each clock cycle. Once the handshake with WE-DATA with the address 0x14 from M1 (i.e., transaction packet length value) is successful, the IP writes this value into TLR. This completes the process of writing the entire transaction packet into the TX-FIFO and writing the corresponding packet length value into TLR.

2. IP→A-S

– Store-and-Forward Mode

The IP compares the number of transaction bytes written to the TX-FIFO with the corresponding value in the TLR. If they match, the transaction can be forwarded to the A-S. Otherwise, a TSE interrupt is generated and output to the Interrupt Controller.

– Cut-Through Mode

If the TX-FIFO is non-empty and the number of transactions is greater than one, part of the transaction can be forwarded to the A-S first. Once the IP writes a valid TLR value and this value matches the number of transaction bytes written to the IP, the remaining transaction of at least one clock cycle can be forwarded to the A-S. Otherwise, a TSE interrupt is generated and output to the Interrupt Controller.

When the interface is set to M2, the IP stores and forwards transactions from M2 to A-S.

1. M2→IP

The IP writes the WE-DATA with the address 0x00 from M2 into the TX-FIFO in each clock cycle. Once the handshake with WE-DATA with the address 0x14 from M1 (i.e., transaction packet length value) is successful, the IP writes this value into TLR. This completes the process of writing the entire transaction packet into the TX-FIFO and writing the corresponding packet length value into TLR.

2. IP→A-S

– Store-and-Forward Mode

The IP compares the number of transaction bytes written to the TX-FIFO with the corresponding value in the TLR. If they match, the transaction can be forwarded to the A-S. Otherwise, a TSE interrupt is generated and output to the Interrupt Controller.

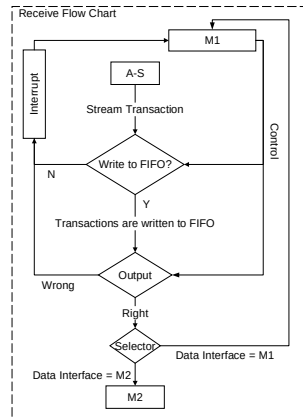
– Cut-Through Mode

If the TX-FIFO is non-empty and the number of transactions is greater than one, part of the transaction can be forwarded to the A-S first. Once the IP writes a valid TLR value and this value matches the number of

transaction bytes written to the IP, the remaining transaction of at least one clock cycle can be forwarded to the A-S. Otherwise, a TSE interrupt is generated and output to the Interrupt Controller.

3.4.2 Operating Principle for Receiving

Figure 3-16 Receive Flow Diagram



The receive flow diagram is as shown in Figure 3-16, and it works as follows:

When the interface is set to M1, the IP stores and forwards a transaction from A-S to M1.

1. A-S→IP

The IP writes the stream transaction from A-S into the RX-FIFO, with last=1 determining the transaction packet boundary, and calculating the RLR value for each transaction.

2. IP→M1

- Store-and-Forward Mode

M1 reads the current RLR value (OE-DATA with address of =0x24), then reads the transaction based on the RLR value (OE-DATA with address of 0x20).

- Cut-Through Mode

Allows M1 to read transaction and RLR value out of order, with the highest bit of RLR determining the transaction packet boundary.

When the interface is set to M2, the IP stores and forwards a transaction from A-S to M2.

1. A-S→IP

The IP writes the stream transaction from A-S into the RX-FIFO, with last=1 determining the transaction packet boundary, and calculating the RLR value for each transaction.

2. IP→M2

- Store-and-Forward Mode

M1 reads the current RLR value (OE-DATA with address of =0x24), then M2 reads the transaction (OE-DATA with address of 0x00) based on the RLR value.

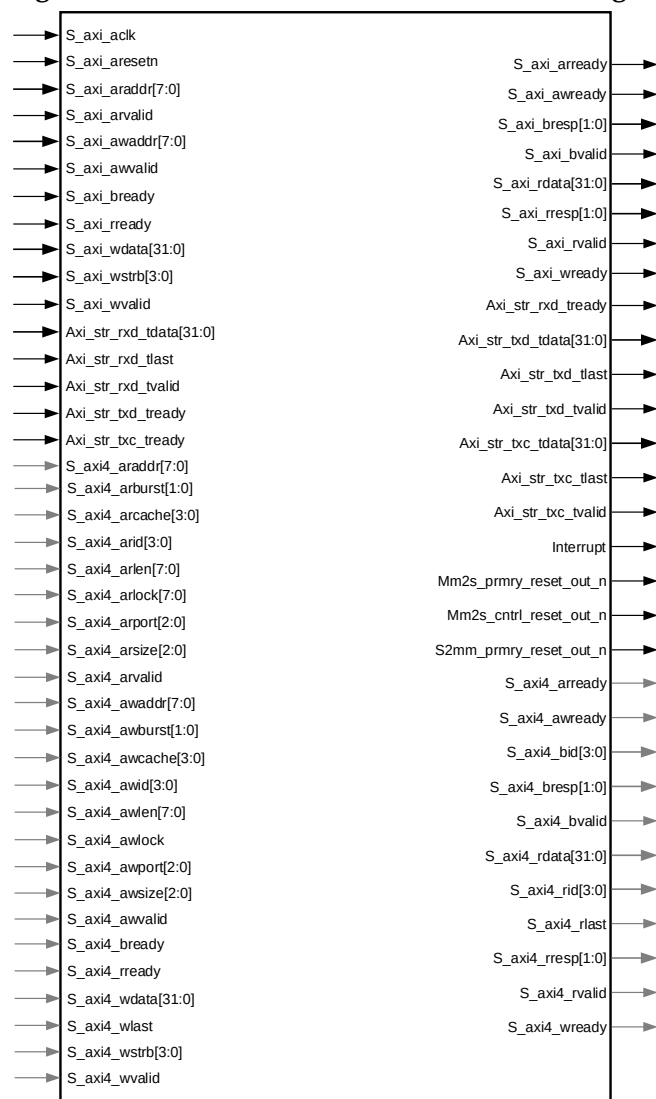
- Cut-Through Mode

Allows M2 and M1 to read transaction and RLR value out of order, with the highest bit of RLR determining the transaction packet boundary.

4 Port Description

The port diagram of Gowin AXI-Stream FIFO IP is shown in Figure 4-1.

Figure 4-1 Gowin AXI-Stream FIFO IP Port Diagram



Note!

Ports with gray arrow will be shown or hidden based on the selected mode.

4.1 System Signal

Table 4-1 System Signal

No.	Signal Name	Type	Description	Data Width	Note
1	S_axi_aclk	Input	Clock	1	Only valid at rising edge
2	S_axi_aresetn	Input	Reset	1	Active-low

Note!

The system signal is input from M1.

4.2 AXI4-Lite Signal

Table 4-2 AXI4-Lite Signal

No.	Signal Name	Type	Description	Data Width	Note
1	S_axi_araddr	Input	Read address	8	–
2	S_axi_arready	Output	Read address ready	1	Active-high
3	S_axi_arvalid	Input	Read address ready	1	Active-high
4	S_axi_awaddr	Input	Write address	8	–
5	S_axi_awready	Output	Write address ready	1	Active-high
6	S_axi_awvalid	Input	Write address ready	1	Active-high
7	S_axi_bresp	Output	Write response	2	00: Valid
8	S_axi_bready	Input	Write response ready	1	Active-high
9	S_axi_bvalid	Output	Write response ready	1	Active-high
10	S_axi_rdata	Output	Read transaction	32	–
11	S_axi_rready	Input	Read transaction ready	1	Active-high
12	S_axi_rresp	Output	Read transaction response	2	00: Valid
13	S_axi_rvalid	Output	Read transaction ready	1	Active-high

No.	Signal Name	Type	Description	Data Width	Note
14	S_axi_wdata	Input	Write transaction	32	–
15	S_axi_wready	Output	Write transaction ready	1	Active-high
16	S_axi_wstrb	Input	Write byte valid	4	The current version only supports all bits set to 1.
17	S_axi_wvalid	Input	Write transaction ready	1	Active-high
18	Interrupt	Output	Interrupt signal	1	Active-high

Note!

These signals are used to connect the IP with M1.

4.3 AXI4 Signal

Table 4-3 AXI4 Port Signal

No.	Signal Name	Type	Description	Data Width	Note
1	S_axi4_araddr	Input	Read address	8	Only 0x00 is enabled in the current version
2	S_axi4_arburst	Input	Read burst	2	It is not enabled in the current version.
3	S_axi4_arcache	Input	Read cache	4	It is not enabled in the current version.
4	S_axi4_arid	Input	Read address ID	4	It is not enabled in the current version.
5	S_axi4_arlen	Input	Read length	8	It is not enabled in the current version.
6	S_axi4_arlock	Input	Read lock	1	It is not enabled in the current version.
7	S_axi4_arport	Input	Read protection	3	It is not enabled in the current version.
8	S_axi4_arready	Output	Read address ready	1	Active-high
9	S_axi4_arsize	Input	Read transaction size	3	It is not enabled in the current version.
10	S_axi4_arvalid	Input	Read address ready	1	Active-high
11	S_axi4_awaddr	Input	Write address	8	Only 0x00 is enabled in the current version
12	S_axi4_awburst	Input	Write burst	2	It is not enabled in the current version.
13	S_axi4_awcache	Input	Write cache	4	It is not enabled in the current version.
14	S_axi4_awid	Input	Write address ID	4	It is not enabled in the

No.	Signal Name	Type	Description	Data Width	Note
					current version.
15	S_axi4_awlen	Input	Write length	8	It is not enabled in the current version.
16	S_axi4_awlock	Input	Write lock	1	It is not enabled in the current version.
17	S_axi4_awport	Input	Write protection	3	It is not enabled in the current version.
18	S_axi4_awready	Output	Write address ready	1	Active-high
19	S_axi4_awsiz	Input	Write transaction size	3	It is not enabled in the current version.
20	S_axi4_awvalid	Input	Write address ready	1	Active-high
21	S_axi4_bid	Output	Write response ID	4	It is not enabled in the current version.
22	S_axi4_bready	Input	Write response ready	1	Active-high
23	S_axi4_bresp	Output	Write response	2	00: Valid
24	S_axi4_bvalid	Output	Write response ready	1	Active-high
25	S_axi4_rdata	Output	Read transaction	32/64	–
26	S_axi4_rid	Output	Read transaction ID	4	It is not enabled in the current version.
27	S_axi4_rlast	Output	Read transaction ends	1	Active-high
28	S_axi4_rready	Input	Read transaction ready	1	Active-high
29	S_axi4_rresp	Output	Read transaction response	2	00: Valid
30	S_axi4_rvalid	Output	Read transaction ready	1	Active-high
31	S_axi4_wdata	Input	Write transaction	32/64	–
32	S_axi4_wlast	Input	Write transaction ends	1	Active-high
33	S_axi4_wready	Output	Write ready	1	Active-high
34	S_axi4_wstrb	Input	Write valid	4/8	The current version only supports all bits set to 1.
35	S_axi4_wvalid	Input	Write ready	1	Active-high

Note!

These signals are used to connect the IP with M2. When The Data Interface is set to AXI-Lite, and all these ports are hidden.

4.4 AXI4-Stream Signal

Table 4-4 AXI4-Stream Signal

No.	Signal Name	Type	Description	Data Width	Note
1	Axi_str_txd_tdata	Input	TX stream transaction	32/64	–
2	Axi_str_txd_tlast	Output	TX ends	1	Active-high
3	Axi_str_txd_tready	Input	TX ready	1	Active-high
4	Axi_str_txd_tvalid	Input	TX ready	1	Active-high
5	Axi_str_txc_tdata	Output	TX transaction monitoring	32/64	It is not enabled in the current version.
6	Axi_str_txc_tlast	Input	TX end monitoring	1	It is not enabled in the current version.
7	Axi_str_txc_tready	Output	TX ready monitoring	1	It is not enabled in the current version.
8	Axi_str_txc_tvalid	Input	TX response monitoring	1	It is not enabled in the current version.
9	Axi_str_rxd_tdata	Input	RX stream transaction	32/64	–
10	Axi_str_rxd_tlast	Input	RX ends	1	Active-high
11	Axi_str_rxd_tready	Output	RX ready	1	Active-high
12	Axi_str_rxd_tvalid	Input	RX ready	1	Active-high
13	Mm2s_prmry_reset_out_n	Output	IP→A-S transaction reset indicator	1	Active-low
14	Mm2s_cntrl_reset_out_n	Output	IP→A-S monitoring reset indicator	1	It is not enabled in the current version.
15	S2mm_prmry_reset_out_n	Output	A-S→IP transaction reset indicator	1	Active-low

Note!

These signals are used to connect IP with A-S.

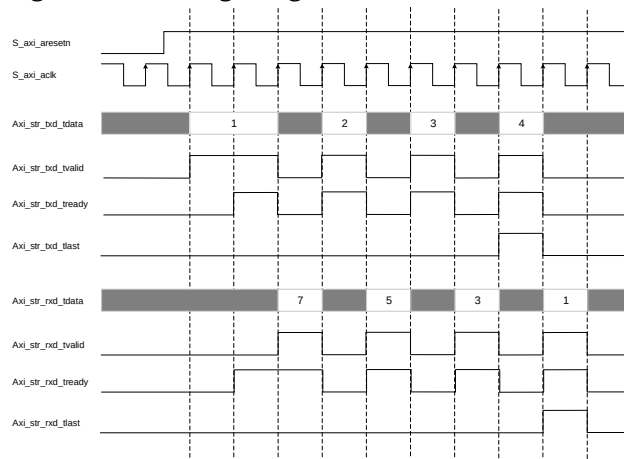
5 Timing Description

Gowin AXI-Stream FIFO IP supports the AXI4 protocol. This manual only introduces the IP interface timing; for detailed information on the AXI4 protocol, please refer to the official standard.

- Valid signals include awvalid, wvalid, bwvalid, arvalid, rvalid, tvalid.
- Ready signals include awready, wready, bready, arready, rready, tready.
- Last signals include Axi_str_rxd_tlast, Axi_str_txd_tlast.

5.1 AXI4-Stream Timing

Figure 5-1 Timing Diagram for AXI4-Stream and IP Interface

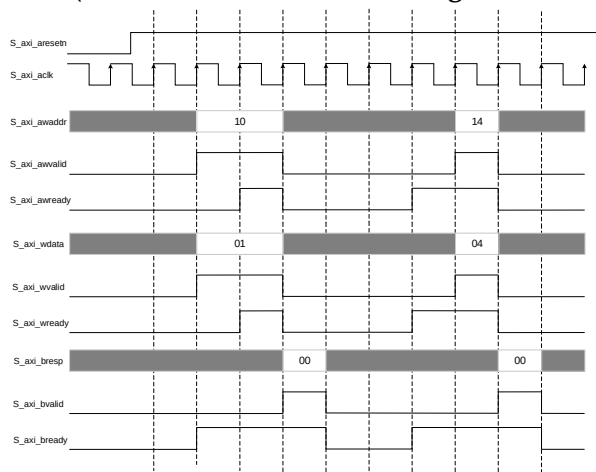


The interface timing for the IP and A-S is as shown in Figure 5-1. The transmit and receive operations are independent. When valid is 1 and ready is 1, it indicates a successful handshake. The last=1 marks the last cycle of each stream transaction.

5.2 AXI4-Lite Timing

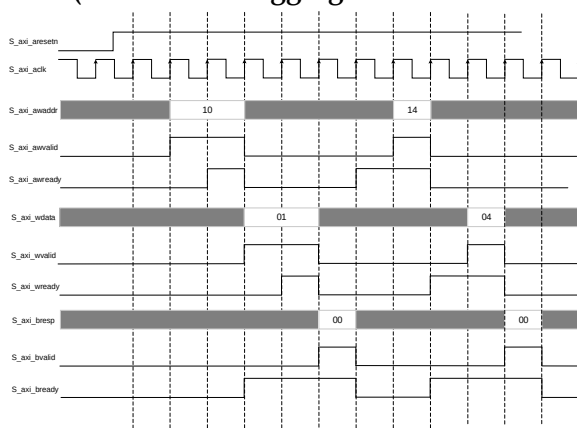
5.2.1 AXI4-Lite Transmit Timing

**Figure 5-2 Write Timing Diagram for M1 and IP Interface
(Address and Transaction Aligned Transfer)**



- For address and transaction aligned transfer initiated by M1 is as shown in Figure 5-2; if the IP is ready to handshake with M1, then awready, wready, and bvalid are set to 1 (otherwise they are set to 0). Concurrently or subsequently, the IP will perform the WE-ADDR, WE-DATA, and WE-RESP handshakes with M1. A handshake is considered successful when ready = 1 and valid = 1, after which awready, wready, and bvalid are set to 0.

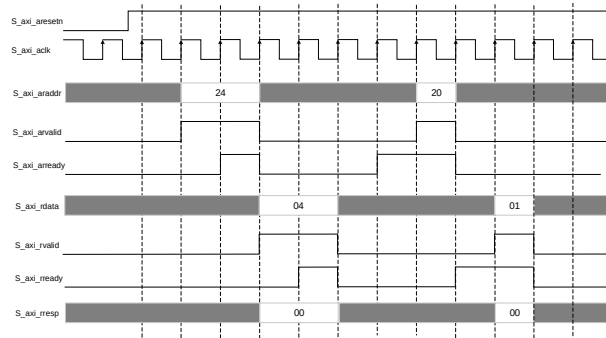
**Figure 5-3 Write Timing Diagram for M1 and IP Interface
(Transaction Lagging behind Address Transfer)**



- For the transaction lagging behind address transfer initiated by M1 is as shown in Figure 5-3; the IP will set awready/wready/bvalid to 1 (otherwise 0) when it is ready to handshake with M1. Subsequently, the IP will perform the WE-ADDR, WE-DATA, and WE-RESP handshakes with M1. A handshake is considered successful when ready = 1 and valid = 1, after which awready/wready/bvalid will be set to 0.

5.2.2 AXI4-Lite Receive Timing

Figure 5-4 Read Timing Diagram for M1 and IP Interface

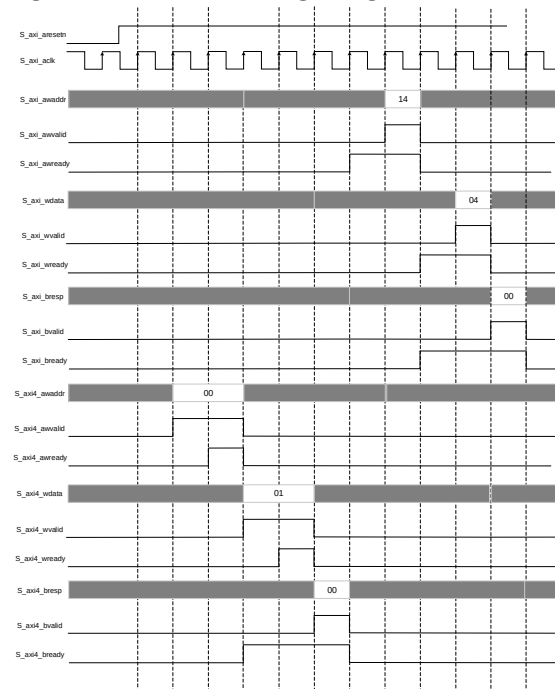


The receive timing is as shown in Figure 5-4. When the IP is ready to handshake with M1, it sets arready/rvalid to 1 (otherwise 0). The IP will then perform the OE-ADDR and OE-DATA handshakes with M1. A successful handshake is indicated by valid=1 and ready=1, after which arready/rvalid will be set to 0.

5.3 AXI4 Timing

5.3.1 AXI4 Transmit Timing

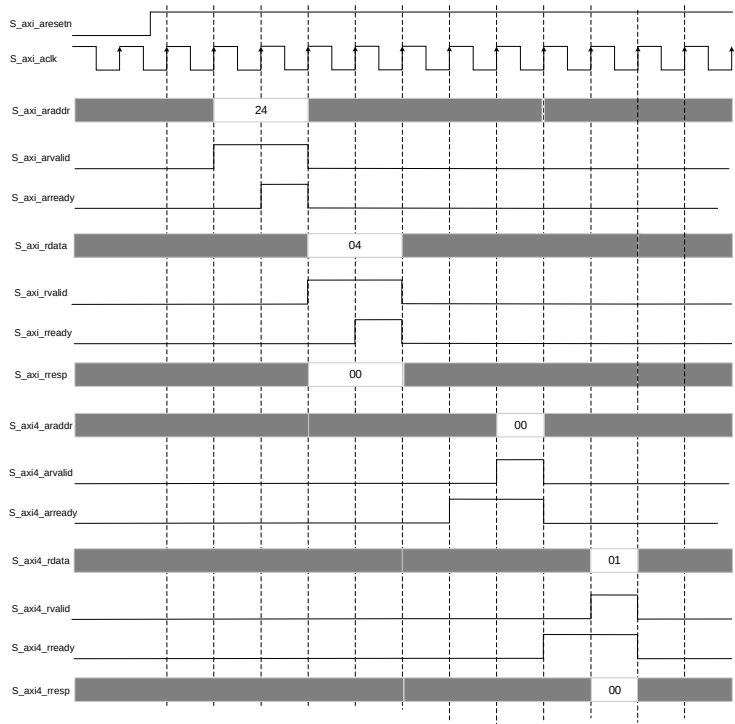
Figure 5-5 Write Timing Diagram for M2 and IP Interface



The AXI4 transmit timing is as shown in Figure 5-5.

5.3.2 AXI4 Receive Timing

Figure 5-6 Read Timing Diagram for M2 and IP Interface



The AXI4 receive timing is as shown in Figure 5-6.

6 Call and Configure IP

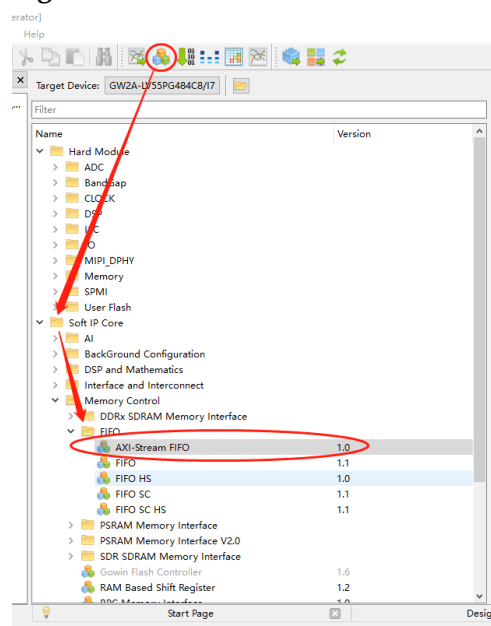
6.1 Call IP

Click " " on the tool bar or "Tools > IP Core Generator" on the menu bar in Gowin Software to call and configure Gowin AXI-Stream FIFO IP.

1. Open IP Core Generator.

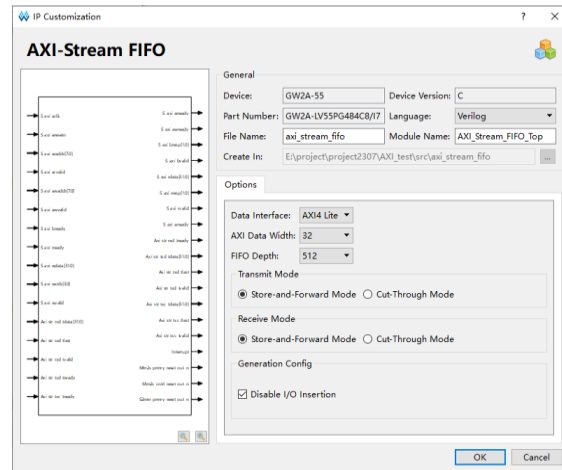
After creating a new project, click "IP Core Generator" to open the IP Core Generator, as shown in Figure 6-1.

Figure 6-1 IP Core Generator Interface



2. Open Gowin AXI-Stream FIFO IP

Double-click "FIFO > AXI-Stream FIFO" to open the AXI-Stream FIFO IP interface, as shown in Figure 6-2.

Figure 6-2 AXI-Stream FIFO Configuration Interface

- You can configure the generated file name in the "File Name" text box.
- You can configure the top module name in the "Module Name" text box.
- You can configure other options by modifying "Options" tab.

6.2 Configure IP

Table 6-1 IP Configuration Option

Option	Description
Data Interface	AXI4 or AXI4-Lite
AXI Date Width	In AXI4 mode, 32/64 bits is available; in AXI-Lite mode, 32 bits is available.
FIFO Depth	TX-FIFO and RX-FIFO depth: 512/1024/2048/4096/8192/16384
Transmit Mode	Store-and-Forward Mode, Cut-Through Mode
Receive Mode	Store-and-Forward Mode, Cut-Through Mode

6.2.1 Data Interface

- When Data Interface is set to AXI4-Lite, the transaction width is 32 bits. TX-FIFO is used to store and forward M1 transactions, and RX-FIFO is used to store and forward A-S transactions.
- When Data Interface is set to AXI4, the transaction width is 32/64 bits. TX-FIFO is used to store and forward M2 transactions, and RX-FIFO is used to store and forward A-S transactions.

6.2.2 Transmit/Receive Mode

- Store-and-Forward Mode: Select the store-and-forward mode to transfer.
- Cut-Through Mode: Select Cut-Through mode to transfer.

For the transfer mechanism of store-and-forward mode and cut-through mode, see [3.3.4 TLR](#) and [3.3.7 RLR](#).

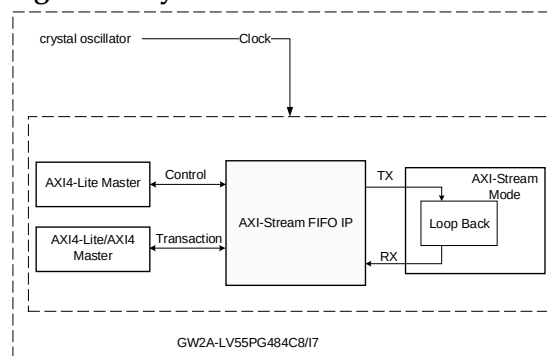
7 Reference Design

For more information, please see the [AXI-Stream FIFO IP](#) reference design at Gowin website.

7.1 Hardware Platform

This section describes the routines for implementing Gowin AXI-Stream FIFO IP on GW2A-LV55PG484C8/I7_V3.0. The system structure is as shown in Figure 7-1. The system consists of the crystal oscillator, M end, IP, and A-S.

Figure 7-1 System Structure of Reference Design



7.2 Operating Principles

The clock is generated by the on-board crystal oscillator, and the working principle is as follows.

1. The M initiates a write transaction, and the IP writes the memory-mapped transaction into the TX-FIFO.
2. M1 initiates the packet length value of a write transaction, and the IP writes the packet length value into the TLR.
3. The IP reads the transactions written into the TX-FIFO based on the value of the TLR and forwards them to the A-S.
4. A-S loops the transaction back to the IP
5. The IP writes the stream transactions into the RX-FIFO, calculates the packet length value, and writes it into the RLR.
6. There are two modes: store-and-forward mode and cut-through mode.

- Store-and-Forward Mode: M1 reads the RLR, and the M end reads the transactions based on the value of the RLR.
- Cut-Through Mode: There is no need to sequentially read the RLR and transactions. The highest bit of the RLR determines the transaction packet boundary.

7.3 Parameter Configuration

1. The on-board crystal oscillator provides a 25 MHz input clock. Users can invoke the PLL to obtain the required clock frequency according to their needs (the current hardware platform design supports a maximum clock frequency of 150 MHz by default).
2. Users can configure parameters in the master_define.v file according to their needs.

7.4 Operating Flow

1. Open Gowin Software (V1.9.9Beta-3 and above) > Open routine project and set top layer file > Compile routine > Download the generated bitstream file (ao.fs) to the board.
2. Capture timing results via Gowin Analyzer Oscilloscope "".
3. Configure master_define.v, and you can refer to the comments to modify parameters as needed.

8 File Delivery

The delivery file of Gowin AXI-Stream FIFO IP includes documentation and reference design.

8.1 Documentation

The folder contains the user guide in PDF version.

Table 8-1 Document List

Name	Description
IPUG1028, Gowin AXI-Stream FIFO IP User Guide	Gowin AXI-Stream FIFO IP User Guide

8.2 Reference Design

Table 8-2 Reference Design File List

File Name	Description
AXI_Stream_FIFO_TB.v	The top-level file of the IP routine, which provides interface information, unencrypted
AXI_Stream_FIFO.v	RTL design file, encrypted
master_define.v	User-configurable parameter file, not encrypted
AXI_test.cst	Physical constraint file, unencrypted (user-configurable)
clk_constraints.sdc	Timing constraint file, unencrypted (user-configurable)

